# Anuj Sharma | Aspiring AI & Data Engineer

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## About

Computer Science and Engineering (Data Science) student at VIT with hands-on experience in Python, SQL, and Machine Learning, building data-driven, AI-enabled applications end-to-end. Strong analytical and statistical problem-solving skills, with practical exposure to vector datastores (FAISS), data preprocessing, and data visualization. Enjoys uncovering actionable insights from data and collaborating with teams to deliver real-world impact.

## Education

**Vellore Institute of Technology, Vellore** **2023 – 2027**
*B.Tech. in Computer Science and Engineering (Data Science) – CGPA: 8.34/10*

**DAV Public School** **2022**
*Senior Secondary (CBSE) – 92%*

**St. Vincent's High and Technical School** **2020**
*Secondary (ICSE) – 85%*

## Technical Skills

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|---|---|
| **Languages:** | Python, SQL, Java, C++, C |
| **Data Science & Analytics:** | Pandas, NumPy, Data Cleaning, Data Preprocessing, EDA, Statistical Modeling & Analysis, Data Visualization |
| **AI/ML:** | Machine Learning, Scikit-learn, Vector Datastores (FAISS), Generative AI, LLMs, RAG, LangChain, LangGraph |
| **Core CS:** | Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks |
| **Backend & Web:** | Spring Boot, REST APIs, FastAPI, Node.js, WebSocket (STOMP), JWT Authentication, React, HTML/CSS |
| **Databases:** | PostgreSQL, MySQL, MongoDB, Redis |
| **DevOps & Tools:** | Docker, Git, Automated Testing (JUnit) |

## Projects

**SkyNetra** | Satellite Image Intelligence Platform GitHub

- Engineered an end-to-end Computer Vision application processing 1,000+ satellite images by integrating Python data pipelines, YOLOv8 object detection, FastAPI REST APIs, and a React interface.
- Optimized image retrieval using FAISS similarity search with ResNet50 embeddings, and applied unsupervised Machine Learning (K-Means clustering) on detection outputs to achieve 87% cluster purity.
- Documented system architecture, API workflow, and processing pipeline to support reproducible development.

**Tech Stack:** Python, FastAPI, REST APIs, React, JavaScript, YOLOv8, FAISS, Git

**SlowMO** | Concurrency-Safe Ticket Booking Platform GitHub | Live Demo

- Engineered a concurrency-safe ticket booking backend in Java Spring Boot 3 with PostgreSQL and Redis, applying pessimistic row-level locking (`SELECT ... FOR UPDATE`) and JPA optimistic versioning to prevent double-booking under simultaneous seat requests.
- Designed 25+ REST endpoints with JWT authentication and RBAC, a Redis-backed seat hold system (10-min TTL auto-release), a FIFO waitlist, real-time WebSocket/STOMP seat-map sync, and ZXing QR-based ticketing; validated concurrency with automated tests (100 concurrent threads → 1 success, 99 correctly rejected).

**Tech Stack:** Java, Spring Boot, PostgreSQL, Redis, WebSocket/STOMP, JWT, Docker, React, Git

**ZenStudent AI** | Real-Time Physiological Signal Analysis GitHub | Live Demo

- Engineered a real-time Machine Learning application processing 5,000+ HRV/EDA samples from 100+ users using statistical data-processing workflows and REST APIs.
- Achieved 93% accuracy on a supervised Machine Learning (classification) model by performing feature engineering on physiological signals and training a Scikit-learn model for automated stress detection.
- Built structured dashboards and data visualizations integrating preprocessing, inference, and backend services to present model results and actionable insights.

**Tech Stack:** Python, Node.js, REST APIs, Scikit-learn, Pandas, NumPy, JavaScript, Git

## Leadership Experience

**Student Coordinator, Riviera Fest** **2026**
*Vellore Institute of Technology*

- Coordinated 20+ cross-functional members to support execution and documentation for Riviera, VIT's flagship fest serving 30,000+ attendees.

**Operations & Finance Executive, E-Cell VIT** **2024 – 2025**
*Vellore Institute of Technology*

- Reduced operational costs by 15% across 10+ events by analyzing budgets, identifying cost drivers, and presenting data-driven recommendations.
- Reduced execution delays by 20% by streamlining workflows for a 15-member team and coordinating logistics for events with 500+ participants.

## Certifications

- Oracle Certified Generative AI Professional – Oracle (2024) – Certificate

## Coursework

Data Structures and Algorithms, Design and Analysis of Algorithms, Database Management Systems, Object-Oriented Programming